



IDENTIFYING FRIEND OR FOE

By Thomas and Titouan



Project Goal

Develop a face detection system capable of:

- Detecting human faces in images
- Checking for the presence of eyes to validate the detection
- Making a decision (fire or deactivate the weapon)
- Providing audio feedback (voice warnings and sound effects)



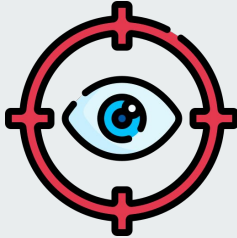
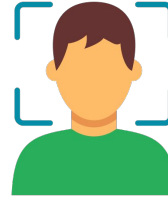
Technologies Used

- Python 3
- OpenCV for image processing and face detection
- Haar Cascade Detection algorithm (frontal and eyes)

Technical Methodology

1. Face Detection

- Uses haarcascade_frontalface_default.xml
- Technique: Haar Features (dark/light patterns on the face)



2. Eye Detection

- Uses haarcascade_eye.xml
- Verifies that there is indeed a human (mutants have no eyes)
- Reduces false positives

3. Decision Logic

- If eyes detected → HUMAN → Disarm
- If no eyes detected → MUTANT → Fire





Program Architecture

- Initialization: Audio configuration, loading classifiers
- Main Loop: Processes each image from the corridor
- Detection: First searches for faces, then for eyes
- Action: Plays a tone (human) or gunshots (mutant)

Project Structure

```
titou@LEGION-TITOUAN:/mnt/d/Cracow_Courses/Python/IdentifyingFriendOrFoe$ tree
.
├── LICENSE
├── Project_9.pdf
├── README.md
├── assets
│   ├── images
│   │   ├── empty_corridor.png
│   │   ├── frame01.PNG
│   │   ├── frame02.png
│   │   ├── frame03.png
│   │   ├── frame04.png
│   │   ├── frame05.png
│   │   ├── frame06.png
│   │   ├── frame07.png
│   │   ├── frame08.png
│   │   └── frame09.png
│   └── sounds
│       ├── gunfire.wav
│       └── tone.wav
├── config.py
├── detects_faces_webcam.py
├── main.py
├── requirement.txt
└── tools
    ├── cascade_classifier.py
    ├── check_requirement.py
    └── tools_image.py

5 directories, 22 files
titou@LEGION-TITOUAN:/mnt/d/Cracow_Courses/Python/IdentifyingFriendOrFoe$
```

Thanks for listening